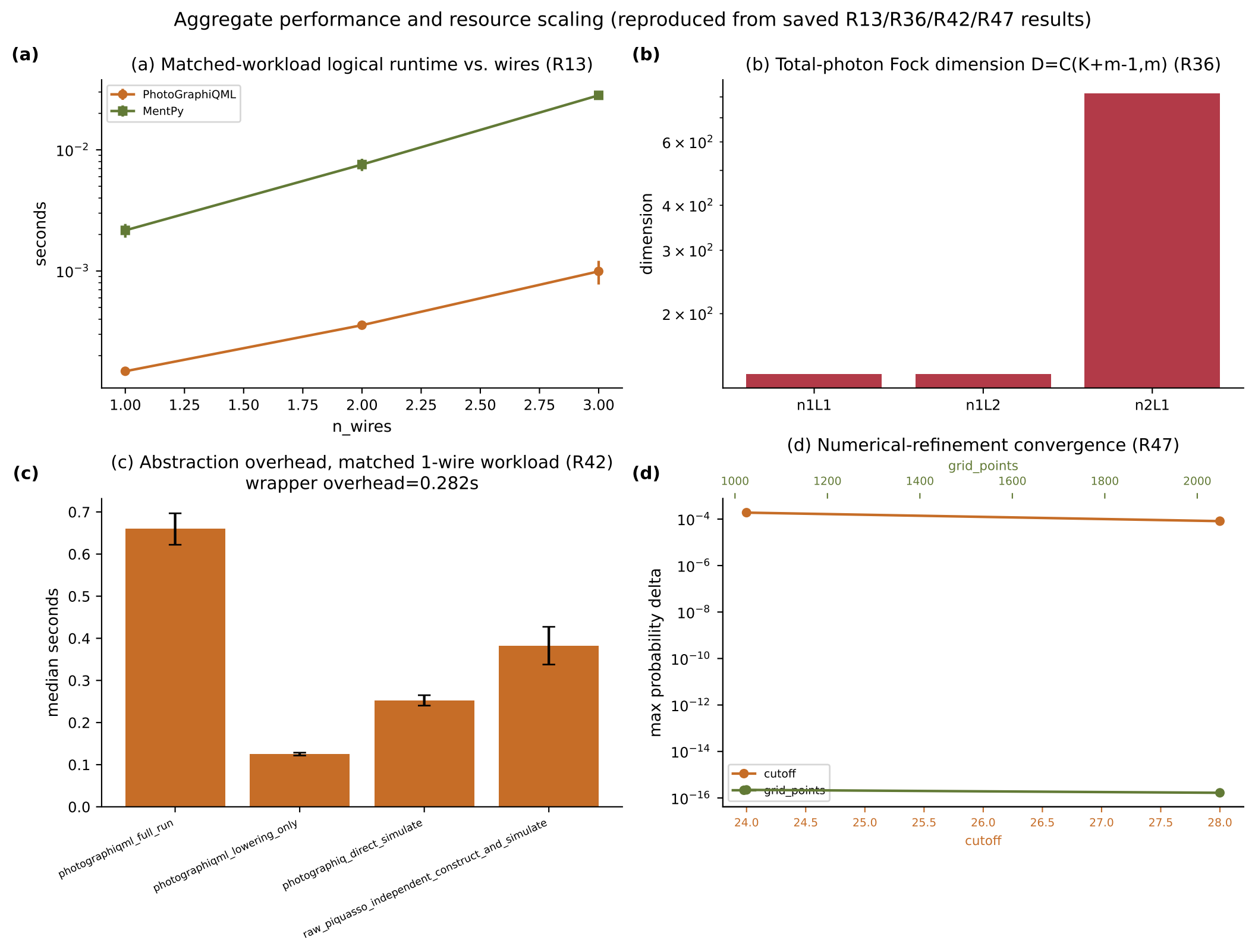
Aggregate performance and resource scaling (reproduced from saved R13/R36/R42/R47 results)
(a)
(a) Matched-workload logical runtime vs. wires (R13)
PhotoGraphiQML
MentPy
seconds
10^{-2}
10^{-3}
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00
n_wires
(b)
(b) Total-photon Fock dimension D=C(K+m-1,m) (R36)
dimension
6×10^2
4×10^2
3×10^2
2×10^2
n1L1
n1L2
n2L1
(c)
(c) Abstraction overhead, matched 1-wire workload (R42)
wrapper overhead=0.282s
median seconds
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
photographiqml_full_run
photographiqml_lowering_only
photographiq_direct_simulate
raw_piquasso_independent_construct_and_simulate
(d)
(d) Numerical-refinement convergence (R47)
grid_points
1000
1200
1400
1600
1800
2000
max probability delta
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}
cutoff
grid_points
24.0
24.5
25.0
25.5
26.0
26.5
27.0
27.5
28.0
cutoff